VeriLoop: Verifiable Digital Service Loops for IoT-Enabled Remote Patient Monitoring

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Abstract

Remote patient monitoring has made it cheap to collect vital signs at home, but most deployments still stop at an alert: nobody checks whether the data behind the alert were trustworthy, whether the action that followed actually helped, or what the system should learn from the outcome. This article argues that connected healthcare needs *verifiable digital service loops* in which sensing, interpretation, decision, execution and validation form one closed and auditable cycle. We present VeriLoop, an edge–fog–cloud architecture that (i) diagnoses sensor faults and confirms suspicious data with the patient before involving clinicians, (ii) maps risk to a tiered service contract bounded by National Early Warning Score 2 (NEWS2) red-flag guardrails, (iii) verifies the physiological outcome of every clinical action and escalates when recovery is not observed, (iv) adapts per-patient operating rules only within clinically approved bounds, and (v) records every loop iteration in a hash-chained provenance ledger anchored in the cloud. In a reproducible simulation of 150 virtual patients monitored at home for one week under five random seeds (750 patient-weeks), with injected sensor faults and connectivity outages, VeriLoop let 5.7% of deterioration episodes reach a critical state, compared with 18.6% for continuous NEWS2 monitoring and 11.7% for an open-loop machine-learning monitor. It also cut false alarms by 38% and 60% respectively, and used 59% less uplink bandwidth than continuous streaming. Removing outcome verification alone nearly doubled the adverse-event rate. We release the simulator and code, and close with the governance and regulatory questions that adaptive, closed-loop health services raise.

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Introduction

Internet of Things (IoT) devices now measure heart rate, respiratory rate, oxygen saturation, blood pressure and temperature at home, and remote patient monitoring (RPM) is routine in chronic and post-discharge care [1], [2]. Yet most RPM deployments are still device-centric: data flow from a wearable to a cloud dashboard, a threshold or early-warning score raises an alert, and the loop ends. Three questions stay unanswered. Was the alert caused by the patient or the sensor? Did the action that followed work? What should the system change for this patient? Without answers, RPM services generate alarm fatigue [3] and cannot produce the evidence that payers and regulators expect.

This special issue frames the alternative as a *digital service loop*: sensing, interpretation, decision, execution and validation, with execution outcomes feeding back into the rules. The idea resembles the MAPE-K loop of autonomic computing [4], but healthcare adds constraints: actions are carried out by people, errors cause clinical harm, adaptation must stay within limits a clinician can sign off, and every step must be auditable. We contribute (i) a definition of a *verifiable* service loop whose validation stage confirms data, verifies outcomes and adapts rules within bounds; (ii) VeriLoop, an edge–fog–cloud realisation with NEWS2 guardrails [5], a hash-chained ledger and an HL7 FHIR mapping [6]; and (iii) an open closed-loop simulator in which we compare VeriLoop with common designs and ablate each stage.

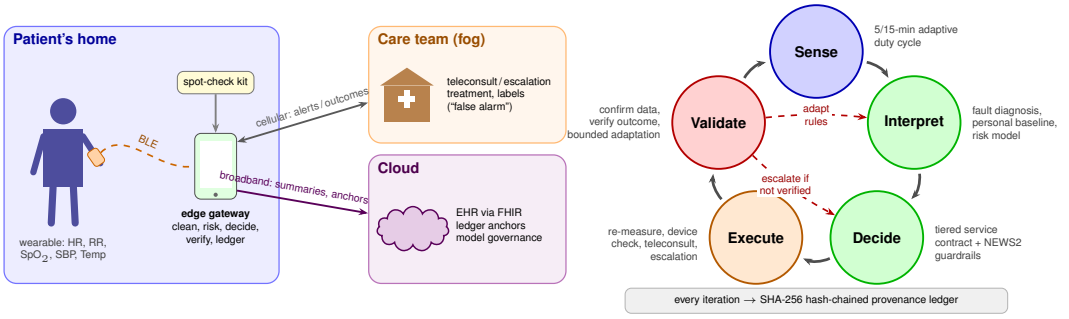
Why Monitoring Pipelines Are Not Loops

IoT health systems typically combine body-worn sensors, a smart edge gateway and cloud analytics [7], [8]; edge processing cuts bandwidth and latency and survives backhaul failures [9].

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The simulator, code and raw results supporting this article are available from the authors (repository link to be added).

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(a) Deployment (b) Verifiable service loop

Fig. 1. VeriLoop at a glance. (a) Deployment: the home edge gateway runs the loop, alerts travel over a cellular fallback, and only summaries and hourly ledger anchors go over the home broadband. (b) The verifiable service loop: validation feeds back into decision (escalate when recovery is not verified) and interpretation (bounded adaptation), and every iteration is written to a hash-chained provenance ledger.

TABLE I. Loop stages closed by common RPM designs (✓ closed, ○ partial, – open).

Stage	Spot NEWS2	Cont. NEWS2	Open-loop ML	**VeriLoop**
Continuous sensing	–	✓	✓	✓
Personalised interpretation	–	–	✓	✓
Fault diagnosis	–	○	○	✓
Data confirmation with patient	–	–	–	✓
Tiered, deadline-tracked actions	○	–	–	✓
Outcome verification	–	–	–	✓
Bounded rule adaptation	–	–	–	✓
Edge operation during outages	✓	○	–	✓
Tamper-evident provenance	–	–	–	✓

NEWS2 remains the clinical reference score: each vital sign scores 0–3, and an aggregate of 5, or any single 3, triggers urgent review [5], [10]. Machine-learning models increasingly sit on top [11]. Seen as a loop, these pipelines are open in three places. **Data are not validated**: signal-quality indices [12] discard bad samples but rarely ask the patient to re-measure or re-seat a probe. **Outcomes are not verified**: after an alert, a fixed cooldown suppresses further alarms, so a failed intervention is noticed only when the cooldown expires. **Rules either never learn or learn without limits**, raising concerns about software that changes after approval [13]. Tamper-evident logging [14], [15] is feasible but rarely built into the loop itself. Table I summarises which loop stages each family of designs closes.

The VeriLoop Architecture

Fig. 1 shows the deployment and the loop. A home edge gateway runs one loop iteration per sample.

Sense. The wearable samples every 15 minutes while the patient is stable and every 5 minutes once risk rises.

Interpret. The edge rejects implausible values, holds back sudden jumps until the next sample confirms them (removing motion spikes), flags frozen channels, and marks an isolated SpO_2 or blood-pressure deviation that no other vital sign corroborates as *suspect*. A logistic-regression risk model turns 16 features (deviation from a 24-hour personal baseline, one-hour mean and 30-minute trend per vital sign, plus NEWS2) into a risk score. We chose an inspectable model whose coefficients a clinician can review.

Decide. A tiered service contract maps risk to actions: Tier 0 routine summaries; Tier 1 patient or device requests (re-measure, re-seat a sensor); Tier 2 teleconsultation when personal risk or NEWS2 reaches amber for two samples; Tier 3 escalation at NEWS2 $\geq$ 7 or very high risk. Every Tier 2 trigger, and every Tier 3 trigger resting on one vital sign, is first confirmed by a Tier 1 re-measurement; an unanswered request escalates anyway. NEWS2 red flags are guardrails that adaptation can never relax.

Execute. Requests are tracked as open service requests with deadlines. Alerts use a cellular fallback, so a broadband outage delays only routine uploads.

Validate. Validation has three jobs. *Confirm the data*: if a re-measurement contradicts the wearable, the trigger is cancelled and a device check is requested. *Verify the outcome*: 90 minutes after treatment starts, VeriLoop escalates if neither risk nor NEWS2 has fallen. *Adapt within bounds*: after a clinician-labelled false alarm, the patient's risk threshold rises by 0.05 (cap 0.85), the NEWS2 amber threshold may rise from 5 to 6, and the baseline is refreshed; this is the kind of pre-declared change envelope regulators ask of adaptive software [13]. Fig. 2 follows one request through these steps.

Provenance and interoperability. Each iteration (observation digest, flags, risk, tier, action) is appended to a SHA-256 hash chain at the edge [14], and the chain head is anchored hourly in the cloud, so edits, reordering or truncation are detectable. Observations map to FHIR `Observation`, requests to `ServiceRequest`

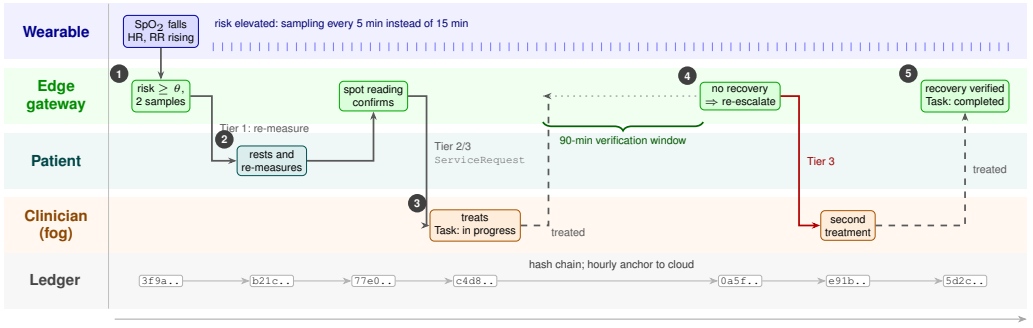


Fig. 2. Lifecycle of one verified service request. (1) The edge detects rising risk and raises the sampling rate. (2) Before calling a clinician it asks the patient to re-measure. (3) The confirmed reading becomes a Tier 2/3 `ServiceRequest`, and the clinician starts treatment. (4) Ninety minutes later no recovery is visible, so the edge escalates again instead of waiting for an alarm cooldown. (5) Recovery after the second treatment is verified and the `Task` is closed. Each step adds a link to the hash chain.

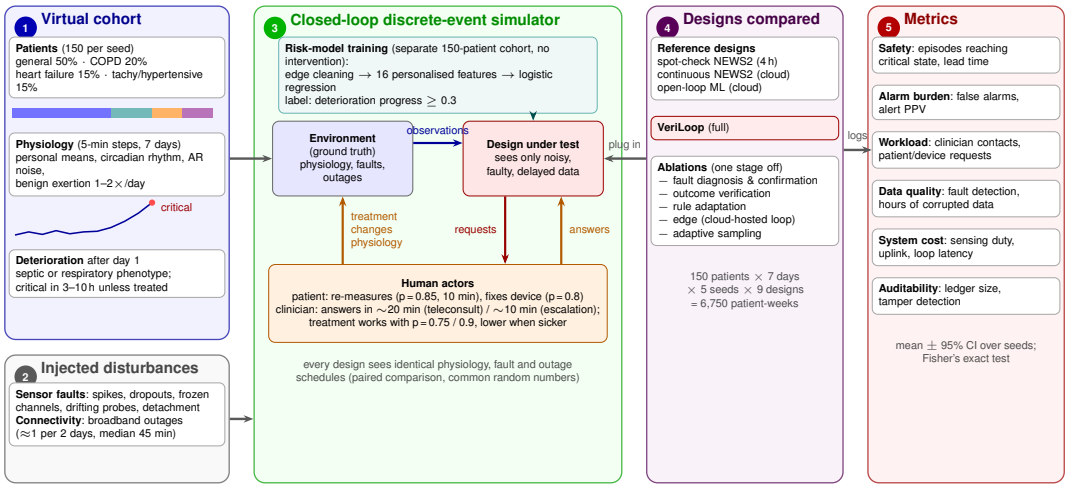


Fig. 3. Evaluation methodology. A virtual cohort (1) and injected sensor faults and outages (2) drive a closed-loop simulator (3) in which the design under test sees only what a real deployment would see, and its requests are answered by simulated patients and clinicians whose treatments change the physiology. Every design (4) runs on identical schedules, and the logs are scored on six groups of metrics (5).

and `Task`, and decisions carry a `Provenance` resource holding the ledger hash [6]. A Tier 2 trigger, for example, becomes a `ServiceRequest` referencing the confirming `Observations`, a `Task` whose status moves from *requested* to *in-progress* when the clinician acts and to *completed* only when the verification stage confirms recovery, and a `Provenance` entry for each transition. Any FHIR-capable electronic health record can therefore display not only that an alert fired, but whether the resulting action was carried out and whether it worked.

Evaluation Methodology

Closed loops need an environment that reacts to alerts, which retrospective datasets cannot provide, so we built a discrete-event simulator (Fig. 3). The results compare designs under identical, controlled conditions and are not estimates of clinical effect sizes.

Patients come from four sub-groups (general, COPD on NEWS2 SpO_2 scale 2, heart failure, tachycardic/hypertensive) with personal baselines, circadian rhythm, noise and daily exertion. About 40% develop septic or respiratory deterioration that becomes critical in 3–10 hours unless effectively treated; reaching that state counts as an adverse event. Devices suffer spikes, dropouts, frozen channels, drifting probes and detachment, and home broadband fails about every two days. Simulated patients answer requests with probability 0.8–0.85; clinicians respond in about 20 minutes (10 for escalations) and their treatment succeeds with probability 0.75–0.9, less often when the patient is sicker.

We compare *spot-check NEWS2* (one reading every 4 hours), *continuous NEWS2* (5-minute streaming, 10-minute persistence, 4-hour cooldown), *open-loop ML* (VeriLoop's risk model in the cloud with a fixed threshold and cooldown) and VeriLoop, plus five ablations. The

TABLE II. Results for all designs (mean over five seeds of 150 patients, 409 deterioration episodes; 95% CI shown for the primary outcome). Lead time runs from the first clinician alert to the point at which the untreated episode would become critical. "Corrupted data" is hours per patient-day with at least one faulty channel; "late" is the share of samples processed more than one minute after sensing.

	Safety		Alarm burden		Workload (/patient-day)		Data quality	System cost		
Design	Critical (%)	Lead (h)	False alarms /patient-day	PPV (%)	Clinician contacts	Patient requests	Corrupted data (h/pd)	Sensing duty (%)	Uplink (KB/pd)	Late (%)
Reference designs										
Spot-check NEWS2 (4 h)	64.5 ± 4.0	1.7	0.37	16	0.50	0.0	4.3	2	0.6	1.8
Continuous NEWS2	18.6 ± 6.0	2.9	0.96	14	1.28	0.0	4.3	100	28.1	1.9
Open-loop ML (cloud)	11.7 ± 3.9	4.6	1.49	11	1.84	0.0	4.3	100	28.1	1.9
Proposed										
VeriLoop	5.7 ± 2.5	4.3	0.59	33	1.07	2.5	2.2	54	11.6	0.0
VeriLoop ablations (one stage removed)										
− fault diagnosis & confirmation	5.7 ± 2.9	4.5	1.68	16	2.20	0.0	4.3	57	13.1	0.0
− outcome verification	10.4 ± 3.2	4.3	0.49	29	0.84	2.2	2.3	54	11.6	0.0
− rule adaptation	6.1 ± 2.3	4.3	0.73	29	1.21	2.7	2.2	55	12.2	0.0
− edge (cloud-hosted loop)	5.5 ± 1.5	4.2	0.59	33	1.09	2.5	2.3	54	15.1	1.7
− adaptive sampling	4.5 ± 2.3	4.3	0.73	28	1.21	3.2	2.0	100	12.2	0.0

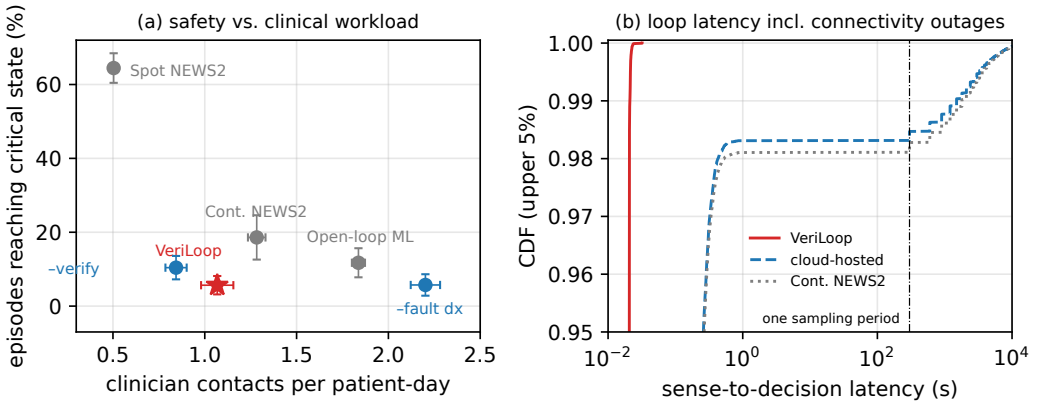


Fig. 4. (a) Share of episodes reaching a critical state versus clinician contacts per patient-day (95% CIs over five seeds). (b) Upper tail of the sense-to-decision latency distribution. Cloud-hosted designs wait for connectivity outages to end; the edge loop does not.

risk model was trained on a separate simulated cohort (validation AUROC 0.99, reflecting the simulator rather than real-world performance). Each design sees the same 150 patients over seven days in each of five seeds.

Metrics. We report, per design, the share of deterioration episodes that reach the critical state (the primary safety outcome), the median lead time between the first alert and that state, false alarms and alert precision, clinician contacts and patient requests per patient-day, hours spent with corrupted data, sensing duty cycle, uplink volume and the share of samples whose sense-to-decision latency exceeds one minute. Confidence intervals are 95% intervals across seeds; differences in adverse events are tested with Fisher's exact test on pooled episodes, and paired seed-level comparisons are used for burden metrics.

Results

Table II summarises all designs; the cohort produced 409 deterioration episodes in 4,500 evaluated patient-days.

Safety and alarm burden. Spot checks missed 28% of episodes and 64% became critical. Continuous NEWS2 caught every episode but late (2.9 hours before the critical point), and 18.6% became critical. Open-loop ML alerted earlier (4.6 hours) and reached 11.7%, but with the most false alarms (1.49 per patient-day). VeriLoop let 5.7% of episodes become critical (23 of 409, against 74 for continuous NEWS2, Fisher's exact $p < 10^{-7}$, and 47 for open-loop ML, $p = 0.004$) while cutting false alarms by 38% and 60%, which doubled alert precision to 33%. It is the only design in the low-risk, moderate-workload corner of Fig. 4(a). The costs: confirmation takes about 18 minutes of lead time relative to open-loop ML, and patients receive 2.5 re-measure or device-check requests per day.

Outcome verification drives most of the safety gain. Without it, the adverse-event rate rose to 10.4% (43 of 409, $p = 0.014$), because a failed treatment was noticed only when the 4-hour cooldown expired. Fig. 5 shows the mechanism on one patient: with verification, the failed

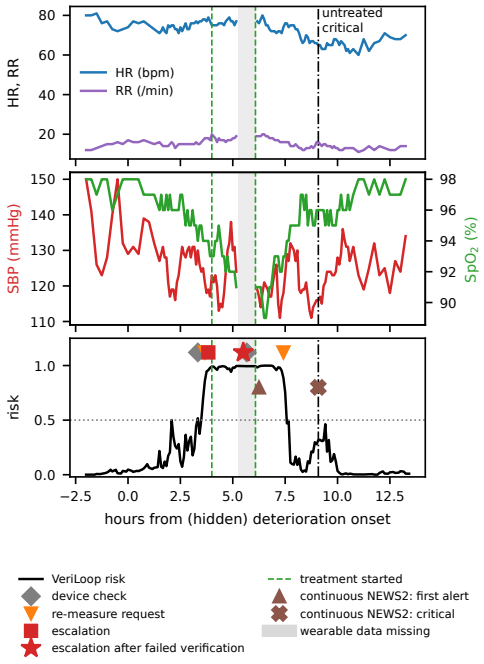


Fig. 5. One simulated loop trace (respiratory deterioration). An early SpO_2 fall is first treated as a suspect reading and checked, then confirmed by re-measurement and escalated. The first treatment fails; outcome verification detects this 90 minutes later and escalates again, and the second treatment works. Under continuous NEWS2 the same patient is first alerted 2.4 hours later and reaches the critical state.

first treatment is caught 90 minutes after it starts, and the second treatment arrives before the critical state. Verification escalations were rare (0.07 per patient-day) and 99% occurred during real episodes, but the full loop needed 0.22 more clinician contacts per patient-day; intervention-specific verification windows would reduce this.

Fault diagnosis halves the noise. Without it, safety was unchanged but false alarms nearly tripled (1.68 per patient-day) and clinician contacts doubled. Device-check requests cut time with corrupted data from 4.3 to 2.2 hours per patient-day. The edge flagged faults with 71% precision and 69% recall (range checks alone: 40% recall); imperfect precision is why suspect readings are re-measured rather than discarded. In Fig. 5 a genuine SpO_2 fall was first treated as suspect, and only the re-measurement prevented a delay.

Adaptation, edge and sampling. Bounded adaptation removed 19% of false alarms with no detectable safety cost. Hosting the loop in the cloud did not change outcomes, because outages are short relative to deterioration, but 1.7% of samples waited over a minute (99th percentile about 16 minutes versus 21 ms at the edge, Fig. 4(b)) and uplink grew 30%. Adaptive sampling halved sensing duty without a significant safety change, and VeriLoop uploaded 11.6 KB per patient-day versus 28.1 KB for streaming.

Auditability is cheap. The ledger stored about 294 entries (44 KB) per patient-day; appending took 1.5 μs and verifying a patient-week 2.5 ms. Verification against the hourly anchors detected all 1,000 random modification, deletion, reordering and truncation attacks; truncation is detectable back to the last anchor.

Lessons and Open Challenges

Validation is a design stage, not a dashboard. The largest gain came from checking whether an action worked, which fits the request lifecycle FHIR `Task` already describes. Every RPM service should state, per action, what recovery is expected and by when. **Adaptation needs a declared envelope**: one-directional, capped steps that never touch red flags turn "a learning system" into a reviewable list of permitted changes, and the ledger proves the envelope was respected [13]. **Loops redistribute work**: confirmation shifts effort from clinicians to patients, so pilots must measure patient burden and adherence.

Deployment considerations. Three practical points follow from building VeriLoop. First, the edge gateway needs only modest resources: the loop logic, the risk model and the ledger together run in under a millisecond per sample, so a home hub or a smartphone suffices, and keeping the loop at the edge also keeps raw vital signs at home unless an action requires them. Second, the tiered contract is a clinical artefact, not a software constant: its tiers, deadlines, verification windows and adaptation bounds should be agreed with the care team, versioned and stored with the ledger so that every decision can be traced to the contract in force at the time. Third, the patient is an actor in the loop, so requests must be clear, few and accessible; patients who cannot re-measure reliably need a carer, a simpler device or a lower Tier 1 threshold for direct clinician review.

Limitations. All results come from simulation with parameterised physiology, faults and human behaviour, and the clinician is assumed to recognise apparent deterioration perfectly. Absolute rates are not clinical predictions. Security was evaluated only for log integrity; confidentiality, device authentication and attacks within an anchor interval remain open.

Open challenges. The community needs shared closed-loop *benchmarks* calibrated on real RPM cohorts, *governance* rules for who is accountable when a patient does not act on a request, and *interoperable service contracts* so tiers, deadlines and verification rules are exchangeable standard resources rather than hard-coded. Beyond these,

closed loops raise new *security* questions: an attacker who can inject a plausible re-measurement or suppress a verification sample can steer the loop, so device attestation and authenticated patient input become part of clinical safety, not only of privacy. Finally, *multi-morbidity* complicates verification, because the expected recovery after an intervention depends on the patient's other conditions and treatments; learning verification windows per patient, within bounds that clinicians approve, is a natural next step for service loops.

Conclusion

Device-centric remote monitoring stops at the alert. VeriLoop closes the loop by validating data before acting, verifying outcomes after acting, adapting only within declared bounds and recording every step in a tamper-evident ledger. In simulation these mechanisms cut the share of deteriorations reaching a critical state by more than two-thirds relative to continuous NEWS2 while lowering false alarms, with outcome verification the most important stage. The next step is calibration on real RPM data and a supervised home-care pilot.

Code and Data Availability

The simulator, VeriLoop implementation, FHIR mapping, experiment scripts and raw results are provided with this article (repository link to be added on acceptance), so that every number in Table II can be regenerated.

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